

Name

- Show appropriate work.
 - Appropriate work for a linear ODE includes the integrating factor and some intermediate steps.
 - Appropriate work for a missing eigenvector includes the linear system you are solving and the solution.
 - Labelling and using eigenvalues/eigenvectors etc. appropriately is a good idea.
- Explicitly real solutions are required.
- Many questions ask only for a linear system. Do not solve the system in these cases.
- Eigenvalues and eigenvectors are given. Do not attempt to recompute them.

Linear ODEs

Section 2.3

1. Solve the ODE $2 y'(t) - 4 y(t) = -6 e^{2t}$.
2. Solve the ODE $2 y'(t) - 4 y(t) = -6 e^{2t}$.
3. Solve the ODE $2 y'(t) - 4 y(t) = -6 e^{2t}$ with $y(0) = 2$
4. Solve the ODE $3 y'(t) - 9 y(t) = 6 e^{3t}$.
5. Solve the ODE $3 y'(t) + 6 y(t) = -3 e^{2t}$ with $y(0) = -3$
6. Solve the ODE $t y'(t) - 3 y(t) = \frac{2}{t^3}$
7. Solve the ODE $t y'(t) - 3 y(t) = -3 t^2$.
8. Solve the ODE $t y'(t) + 2 y(t) = 2 t^2$ with $y(1) = 2$
9. Solve the ODE $t y'(t) + 2 y(t) = 2 t^2$.
10. Solve the ODE $t y'(t) + 3 y(t) = 3 t^3$ with $y(1) = -1$

All with a box saying

$y(t) =$

Complementary Solutions from Eigenvalues/Eigenvectors

Sections 3.3, 3.4, & 3.5

Any and all of the questions from these sections can include. "Discuss the behavior as $t \rightarrow \infty$."

1. Compute the complementary solution of $y' = A y$ where A is below

$$\text{MatrixForm}\left[\text{Eigensystem}\left[A = \begin{pmatrix} -5 & 6 & -9 \\ -3 & 4 & -9 \\ 0 & 0 & -2 \end{pmatrix}\right]\right]$$

Out[81]//MatrixForm=

$$\begin{pmatrix} -2 & -2 & 1 \\ \{-3, 0, 1\} & \{2, 1, 0\} & \{1, 1, 0\} \end{pmatrix}$$

$$y_c(t) =$$

2. Compute the complementary solution of $y' = A y$ where A is below

$$\text{MatrixForm}\left[\text{Eigensystem}\left[A = \begin{pmatrix} 3 & 0 & -5 \\ 5 & -2 & -5 \\ 0 & 0 & -2 \end{pmatrix}\right]\right]$$

Out[82]//MatrixForm=

$$\begin{pmatrix} 3 & -2 & -2 \\ \{1, 1, 0\} & \{1, 0, 1\} & \{0, 1, 0\} \end{pmatrix}$$

$$y_c(t) =$$

3. Compute the complementary solution of $y' = A y$ where A is below

$$\text{MatrixForm}\left[\text{Eigensystem}\left[A = \begin{pmatrix} 4 & 7 & -12 \\ 4 & 8 & -8 \\ 2 & 5 & -6 \end{pmatrix}\right]\right]$$

Out[101]//MatrixForm=

$$\begin{pmatrix} 6 & -2 & 2 \\ \{1, 2, 1\} & \{2, 0, 1\} & \{-1, 2, 1\} \end{pmatrix}$$

$$y_c(t) =$$

4. Compute the complementary solution of $y' = A y$ where A is below

MatrixForm[Eigensystem[A = $\begin{pmatrix} 4 & 5 & -12 \\ 4 & 8 & -8 \\ 2 & 5 & -6 \end{pmatrix}$]]]

Out[105]//MatrixForm=

$$\begin{pmatrix} 4 + 2i & 4 - 2i & -2 \\ \{i, 2, 1\} & \{-i, 2, 1\} & \{2, 0, 1\} \end{pmatrix}$$

$$y_c(t) =$$

5. Compute the complementary solution of $y' = A y$ where A is below

In[154]:=

MatrixForm[Eigensystem[A = $\begin{pmatrix} -6 & -5 & 8 \\ 4 & -2 & -8 \\ 2 & 0 & -6 \end{pmatrix}$]]]

Out[154]//MatrixForm=

$$\begin{pmatrix} -6 + 2i & -6 - 2i & -2 \\ \{i, 2, 1\} & \{-i, 2, 1\} & \{2, 0, 1\} \end{pmatrix}$$

$$y_c(t) =$$

6. Compute the complementary solution of $y' = A y$ where A is below

In[155]:=

MatrixForm[Eigensystem[A = $\begin{pmatrix} 5 & -1 \\ 4 & 1 \end{pmatrix}$]]]

Out[155]//MatrixForm=

$$\begin{pmatrix} 3 & 3 \\ \{1, 2\} & \{0, 0\} \end{pmatrix}$$

$$y_c(t) =$$

7. Compute the complementary solution of $y' = A y$ where A is below

In[185]:=

MatrixForm[Eigensystem[A = $\begin{pmatrix} 5 & -1 \\ 1 & 3 \end{pmatrix}$]]]

Out[185]//MatrixForm=

$$\begin{pmatrix} 4 & 4 \\ \{1, 1\} & \{0, 0\} \end{pmatrix}$$

$$y_c(t) =$$

Undetermined Coefficients

Section 3.6

1. For the ODE $y' = Ay + b$ with $b = e^{3t} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

1.1. Write down the form for the particular solution

$$y_p(t) =$$

1.2. Write down the Lin System for the coefficients

$$$$

2. For the ODE $y' = Ay + b$ with $b = \cos(4t) \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \sin(4t) \begin{pmatrix} 4 \\ 2 \end{pmatrix}$

2.1. Write down the form for the particular solution

$$y_p(t) =$$

2.2. Write down the Lin System for the coefficients

$$$$

3. For the ODE $y' = Ay + b$ with $b = \begin{pmatrix} t^2 \\ 1 \\ 2 \end{pmatrix}$

3.1. Write down the form for the particular solution

$$y_p(t) =$$

3.2. Write down the Lin System for the coefficients

4. For the ODE $y' = Ay + b$ with $b = \begin{pmatrix} e^{2t} \\ -e^{2t} \\ 3e^{2t} \end{pmatrix}$

4.1. Write down the form for the particular solution

$y_p(t) =$

4.2. Write down the Lin System for the coefficients

5. For the ODE $y' = Ay + b$ with $b = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

5.1. Write down the form for the particular solution

$y_p(t) =$

5.2. Write down the Lin System for the coefficients